



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani

Pilani Campus

AUGS/ AGSR Division

FIRST SEMESTER 2025-2026

Course Handout (Part II)

Date: 30/07/2026

In addition to part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. : **MF F315**
Course Title : **AUTOMATION AND CONTROL**
Instructor-in-charge : **PRATEEK KALA**

1. Scope and Objective of the Course:

This course is intended to a comprehensive knowledge of the technology related to Mechatronics and Automation. The necessity of integrating and embedding electronics and microprocessor into mechanical systems have been long felt, due to rapid progress in microprocessor computer based technology, in domestic products to manufacturing systems. Mechatronics is a recently defined engineering field that builds on the traditional mechanical engineering studies, combines it with technologies from the electrical, electronics, computer and control fields, using techniques such as simultaneous engineering to provide solutions in manufacturing applications. Also, mechatronics has been applied to manufacturing and other industrial automation: robotic automation found in car automated production lines, such as welding, and assembly line in computer integrated manufacture etc. Thus, the manufacturing system is greatly enhanced by the application of mechatronics.

This course will develop overall background of the student in interdisciplinary mechatronic technology and a broad introduction to the issues encountered and techniques required in developing mechatronic products and automation systems.

2. Text Book: Bolton W., *Mechatronics*, 3rd Ed., Pearson, 2004. [1]

Reference Books:

- (i) M.P. Groover, “Automation, Production systems, and Computer-Integrated Manufacturing”, PHI, 2008. [2]
- (ii) Anthony Esposito, Fluid power with applications, Pearson, 2008. [3]

3. Lecture Plan

Lect. No.	Topic(s)	Book-Chapter
1	Introduction to scenario in Automation	1-1
2	Automation principles	2-3
3-4	Microcontroller programming	Class notes
5-6	Microcontroller and sensors communication	Class notes
7-8	Microcontroller and actuators communication	Class notes
9	Microcontroller and display	Class notes
10	Sensors: Static and dynamic characteristics	1-2, Class notes
11-12	Sensor selection and application in industry	1-2, class notes
13-14	Signal conditioning: time and frequency domain concepts, OPAMP, filters, DAC/ADC, MUX	1-3, class notes
15	Actuators- Electric	1-9, class notes
16-20	Actuator – hydraulic. Design, circuit analysis	1-7, 3-4, 5, 7 class notes
21-22	Electro pneumatics	Class notes
23-25	Programmable logic controller: Introduction and programming	1-21, class notes
26	Microprocessor and microcontroller	1-17
27	Digital electronics, Digital logic	1-5
28	Communication	Class notes, 1-22
29	Industrial automation scenario	class notes



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30-32	Numerical control machine tools, numerical control part programming, Tool Monitoring System	Class notes
33-35	Basic control concepts: open, close, proportional derivative, integral, multivariable, digital, adaptive control systems	1-15, class notes
36-38	Industrial robots; basics and application, man-machine Interface	class notes
39-42	Automated material handling systems, Automatic identification and data capture, Automated production lines & assembly systems	2-10, 2-12, 2-16

4. **Assignments:**

There will be project, class assignments, home assignments, and case studies. Each student has to work out the given assignments & work on a problem assigned.

5. **Evaluation:**

<u>Component</u>	<u>Duration</u>	<u>Weightage</u>	<u>Date & Time Venue</u>	<u>Remarks</u>
Tests	TBA	25%	TBA	
Lab	2 hrs	30%	Continuous	
Mid Sem	1 ½ hr	20%	Refer Time table	
Compree	3 hrs.	25%	Refer Time Table	

- Component weightages may vary but it will be pre announced.(Subjected to success of implementation of method)
- Makeup request will not be entertained unless prior permission is taken. Students are expected to attend regular class and practical.

6. **Chamber Consultation Hour:** 12-1:15 daily.

7. **Notices:** All notices will be communicated via email and in Nalanda portal.

MF F315
Instructor-in-charge



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